

Topology Optimization of Motor Stator to Reduce Vibration

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Abstract. This paper provides a topology optimization (TO) of motor stator using an on/off method with considering vibration. The on/off method using evolutionary optimization tends to yield complex shapes, which are impossible to produce in view of engineering. To solve this problem, two methods are introduced: a on/off method with filtering and a normalized Gaussian function network (NGnet) method. Optimization results show that the effectiveness of these methods is confirmed in a TO of an interior permanent magnet motor.

Keywords: Electromagnetic-structural coupled analysis, Motor, Topology optimization

1. Introduction

In recent years, as electric motors have become smaller and lighter, higher power density has been required. In conventional motor shapes, vibration and noise are expected to increase due to the increased electromagnetic force per unit volume. The approaches to reduce the vibration are generally classified into two methods: drive strategy and geometric strategy. The former approach is to prevent resonance by adjusting the excitation method. The latter approach is to find a motor shape that can avoid resonance frequencies and associated mode shapes excited by harmonic magnetic forces. This study adapts the latter approach and introduces structural optimization to explore the new stator shapes that reduces vibration.

Structural optimization is generally classified into three types: size (parameter), shape, and topology [1]. The size optimization is a method to discover the optimal design by adjusting size parameters and usually employed in the detailed design process stage. However, it is not guaranteed that the result of the size optimization brings a truly optimal structure, since the shape and topology of the structure are not changed. The shape optimization changes the surface but keeps the topology fixed. It is usually used in the preliminary design stage to select the optimal

shape of the external boundary surface or curve [2]. From the above, the size optimization and the shape optimization have a limitation in the degree of freedom of optimization. To overcome this limitation and explore a novel motor shape, topology optimization (TO) is employed in this study.

TO is usually employed for initial conceptual design stage. Most of the research using TO can be categorized into four main methods: homogenization, evolutionary, density (solid isotropic material with penalization: SIMP) and level set. In the homogenization method, a certain region of the target structure is composed of periodic micro-structures [3]. It can be optimized to take the anisotropy of the material into account. The design variables for the homogenization method are the length, width, and slope of the rectangular pores in the microstructure of each element, and the number of design variables is three times the number of elements. The evolutionary structural optimization achieves optimal material distribution by removing inefficient materials from the structure or by moving them from the strongest to the weakest parts in the structure [4]. The density method involves ascribing a variable density to each element and imposes an artificial penalty on the material properties according to the objective function [5]. The design variable is the material density that determines the material properties of each element. The level set method defines the interface between material phases by contour lines of the level set function [6]. The level set functions given to the nodes of each element are the design variables.

Since these TO deal with a large number of real-valued design variables, gradient-based methods with sensitivity analysis are generally employed. In this method, the calculation cost for sensitivity analysis does not depend on the number of design variables because an adjoint variable method can be employed in most cases. However, the gradient-based method tends to converge to one of the locally optimal solutions if the initial shape is not appropriate. Therefore, it is important for this method to set suitable initial shapes. The most methods without using the gradient-based method has no guarantee that the topology changes to the gradient direction of objective function during optimization, and the computational cost of TO is huge. However, this method tends to obtain an optimal solution independent of the initial shape and does not require the derivative of the objective function.

According to the above categorization of TO, a density method with non-gradient methods is adopted in this study. In general, the density method is an optimization problem that deals with a huge number of design variables that vary continuously from 0 (void) to 1 (solid), thus, it is difficult to obtain an optimal solution using the non-gradient method. In addition, elements with middle properties between air and material (grayscale elements) appear in the density method. It is difficult to treat the grayscale material in TO of motors made by steel. Therefore, in this study, we simplify the optimization problem and avoid the appearance of grayscale elements by treating the continuously varying design variables as binary design variables of 0 or 1. This is called the on/off method [7-9]. The on/off method has a problem that it tends to yield

complex shapes which are impossible to produce in view of engineering. Moreover, the check-board structures significantly degrades the performance of TO with considering vibration, because these structures have poor mechanical strength.

The TO with considering vibration usually follow some general objectives. The first objective is to maximize the specified structural eigenvalues (natural frequency). The second is to maximize the distance between a specified structural eigenvalues and a given frequency. The optimization can be done with respect to free vibration or forced vibration.

To explore a new motor shape, in this study, two TO methods are introduced to improve the performance of TO with considering vibration: an on/off method with filtering and a normalized Gaussian function network (NGnet) method [10-12]. The advantage of the on/off method is easy to apply filtering operators that can remove holes and obtain smooth surces. The NGnet can express smooth as well as complex shapes in contrast with conventional Gaussian network. Therefore, this method tends to obtain a coherent shape and is suitable for TO with considering vibration. The effectiveness of these methods is confirmed in a TO of an interior permanent magnet motor.

2. Formulations

2.1. Electromagnetic-structural coupled analysis

In this study, In order to evaluate the performance of the motor with considering vibration, an electromagnetic-structure coupled analysis using the finite element method is employed. The frequency response equation with n -DoFs in general undamped system can be written as follows:

$$(\mathbf{K} - \omega^2 \mathbf{M})\vec{x} = \vec{f} \quad (1)$$

where $\mathbf{K}, \mathbf{M} \in \mathbf{R}^{n \times n}$ are mass matrix and stiffness matrix, $\vec{x}, \vec{f} \in \mathbf{R}^n$ are nodal displacement and electromagnetic nodal force, respectively. Using a standard mode-superposition technique (mode-displacements method), (1) can be written as,

$$(\Phi^T \mathbf{K} \Phi - \omega^2 \Phi^T \mathbf{M} \Phi)\vec{\delta} = \Phi^T \vec{f} \quad (2)$$

where $\Phi \in \mathbf{R}^{n \times m}$ is modal matrix made by m eigenvectors and $\vec{\delta} \in \mathbf{R}^m$ is displacement in the modal coordinates. The frequency domain vector $\vec{f}$ in (1) and (2) is calculated by the Fourier transformed time domain electromagnetic force that is calculated by using electromagnetic field analysis with nodal force method.

2.2. Topology optimization

Two topology optimization methods are introduced in this study: the on/off method using immune algorithm (IA) and the NGnet method using real-coded genetic algorithm (RGA). In the on/off method, materials are expressed by a 0/1 binary map which corresponding the mesh of finite element method. The IA is employed to optimize the 0/1 binary map and treats this map as an individual. In this study, the on-state material means iron and the off-state material means air. The IA is an evolutionary algorithm inspired by the immune system of living organisms and has both global and local search characteristics. The procedure of the IA is as follows:

1. Initial individuals are generated. The design variables representing each individual are randomly set.
2. The filtering operator is applied to individuals, and then the individuals are evaluated by objective function that is obtained by electromagnetic-structural coupled analysis and ranked.
3. Clones of individuals are generated. The clones corresponding to higher-ranked individuals are more generated.
4. Small-modification on their surface by maturing operator is applied to the clones, and then the filtering operator is applied again.
5. The clones are evaluated by objective function, and replace the original by the clone if the best clone is better than the original (parent).
6. The new individuals are ranked. A few percent of low ranked individuals are culled and the same number of new individuals are added.

To obtain feasible design, we introduced filtering operators that consists of a surface smoothing, hole eliminating operation [10].

The procedure of the surface smoothing and the hole eliminating operation is as follows:

1. List up all air elements that have 2 or 3 edges of material boundary in the design area.
2. Convert material state to iron for all elements in the list.
3. Repeat step 1 and 2 until all air elements are unchanged, and then do the same procedures for all iron elements.

In this study, a floating material removing operation is introduced because the floating material affect the calculation of vibration analysis. The procedure of the floating material removing operation is simple. In this operation, continuous iron blocks are listed up and erased other than the largest blocks. Figure 1 shows examples of the filtering operators.

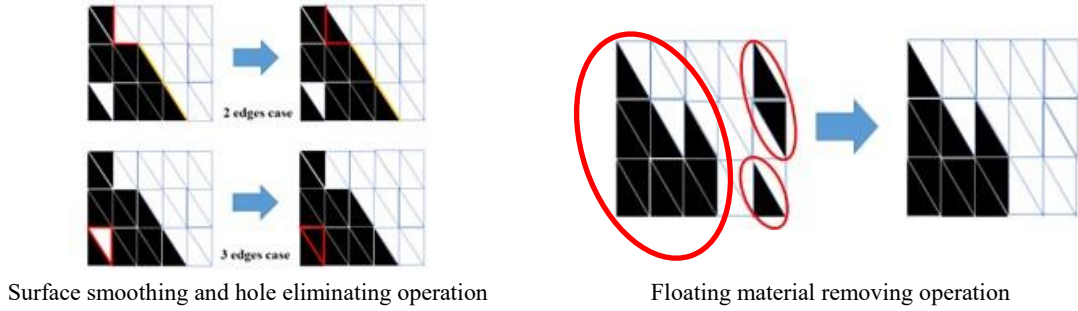


Figure 1: Examples of filtering operators

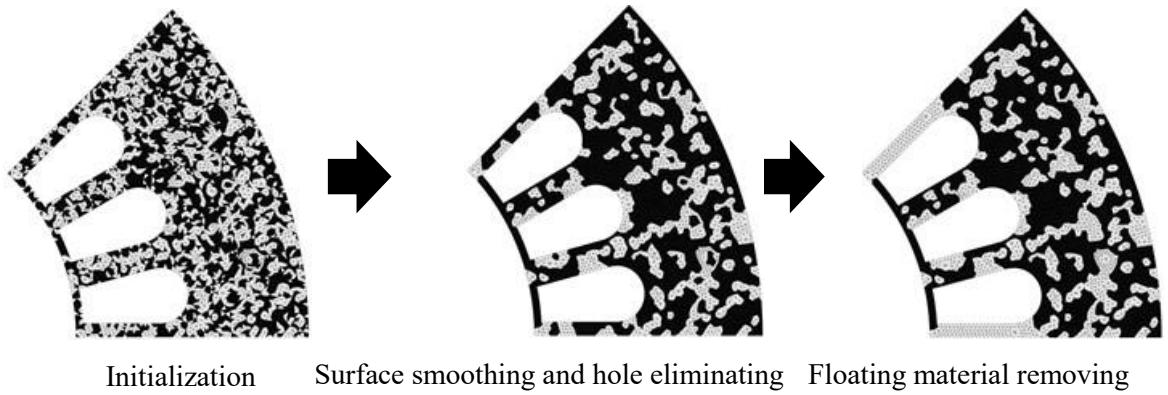


Figure 2: View of filtering effect on individuals after initialization

Figure 2 shows the effect of filtering operator and suggests that the stiffness of design can be improved by the filtering without evaluation process. Hence, the filtering operator increases the efficiency of TO with considering vibration.

In the NGnet method, the optimal shapes are expressed by the NGnet in which the shape function is defined as a weighted linear combination of normalized Gaussians distributed in the design region as shown in Fig. 3[11, 12]. The output of NGnet is as follows:

$$y(\vec{x}_e) = \sum_{i=1}^N w_i b_i(\vec{x}_e) \quad (3)$$

$$b_i(\vec{x}_e) = G_i(\vec{x}_e) / \sum_{k=1}^N G_k(\vec{x}_e) \quad (4)$$

$$G_k(\vec{x}_e) = \frac{1}{(2\pi)^{D/2} |\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2} (\vec{x}_e - \vec{\mu}_k)^T \Sigma_k^{-1} (\vec{x}_e - \vec{\mu}_k)\right\} \quad (5)$$

where $y(\vec{x}_e)$, $\vec{x}_e$, N , w_i , μ_k and Σ_k are output, central vector of element, number of the Gaussian functions, weight coefficient, central vector of the Gaussian function and covariance matrix of the Gaussian function, respectively. For the output of NGnet for each element, the shape function $v(\vec{x}_e)$ is given by using the following step function and attributes the NGnet method to the on/off method.

$$v(\vec{x}_e) = \begin{cases} on & y(\vec{x}_e) \geq 0 \\ off & y(\vec{x}_e) < 0 \end{cases} \quad (6)$$

In order to change $y(\vec{x}_e)$, we need to optimize w_i , μ_k , and Σ_k . However, if all these parameters are to be changed, the cost of optimization is huge. In the general NGnet method used for optimization, w_i are optimized under the fixed μ_k and Σ_k . A real genetic algorithm (RGA) is introduced to optimize w_i in this study. Blend crossover (BLX- α) is employed as crossing operator and just generation gap (JGG) is employed as selection model in the RGA [4, 5].

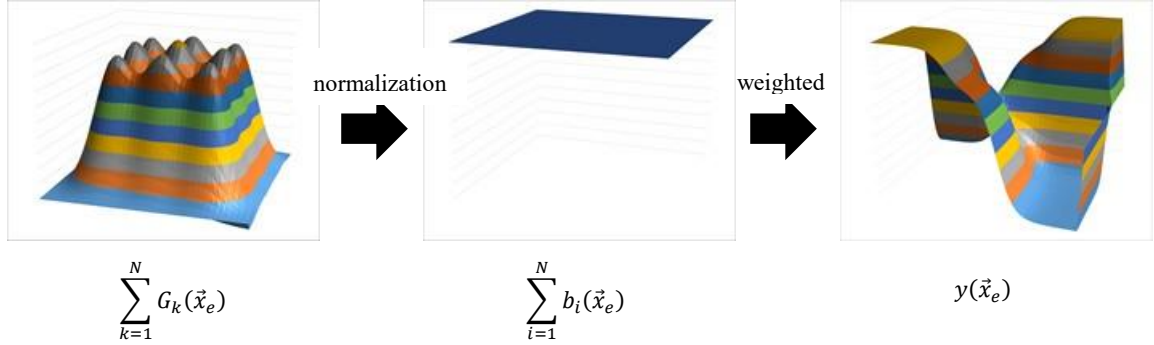


Figure 3: Overview of NGnet.

3. Optimization results

We performed a TO of stator in an interior permanent magnet motor to minimize the surface displacement of the stator while maintaining the torque characteristics. The objective function F includes penalty functions is defined by

$$F = X/X_0 + K \times P_1 + K \times P_2 \rightarrow \min \quad (7)$$

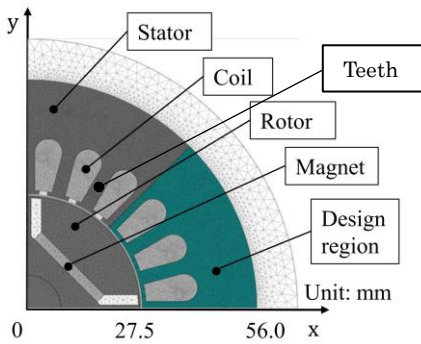
$$P_1 = \begin{cases} 0.0 & (T_{ave} \geq T_{ave0}) \\ (T_{ave0} - T_{ave})/T_{ave0} & (T_{ave} < T_{ave0}) \end{cases}, \quad P_2 = \begin{cases} 0.0 & (T_{rip} \leq T_{rip0}) \\ (T_{rip} - T_{rip0})/T_{rip0} & (T_{rip} > T_{rip0}) \end{cases} \quad (8)$$

where X , X_0 are the average displacement in all nodes of the surface of the stator and its in reference model, T_{ave} , T_{rip} are the average and ripple of torque, T_{ave0} , T_{rip0} are the average and ripple of torque in reference model, respectively. K is penalty coefficient and $K =$

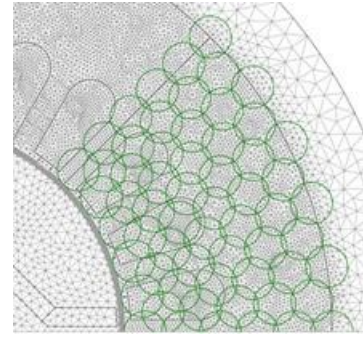
10 in this study. The reference model and distribution of NGnet on design region are shown in Fig. 4. The specification of reference model is summarized in Table 1.

Table 1: The specification of reference model.

Power frequency	100 Hz	Young's modulus of magnetic material	210 Gpa
Rotation speed	3000 rpm	Density of magnetic material	7650 kg/m ³
Number of poles	4	Poisson's ratio of magnetic material	0.3
Number of turns	35 /slot	Young's modulus of magnet	160 Gpa
Current	3 A	Density of magnet	6000 kg/m ³
Current phase	18 deg	Poisson's ratio of magnet	0.2



(a) Reference model



(b) Distribution of NGnet on design region

Figure 4: Reference model and Distribution of NGnet on design region

Figure 5 shows the optimized shape obtained by the present methods. The optimal shape by the on/off method with filtering as shown in Fig. 5 (a) have some voids but no floating material is found. The optimal shape by the NGnet method is coherent shape and no voids is found in Fig 2(b). In both optimal shapes, teeth are connected each other to increase mechanical strength. A comparison of performance values is shown in Table 2. We can see that both methods reduce the average displacement while maintaining the same torque characteristics as the reference model. In particular, the on/off method with filtering reduces the average displacement by about 83%.

We analyze the natural frequencies in both shapes to investigate the reason of vibration reduction. In general, structure with high natural frequency is hard to resonate. A comparison of natural frequencies are shown in Fig. 6. It seems that two optimized shapes have higher overall natural frequencies than that in the reference shapes. Such a large change in vibration characteristics in optimized shapes results from the effect of the filling in between the teeth. Almost natural frequencies of the shape optimized by NGnet method tends to be higher than that by the on/off method because of no bubbles. Electromagnetic force is investigated to find out the

reason why the average displacement is worse than that by on/off method, even though the shape optimized by NGnet is structurally superior. Figure 7 shows a comparison of time domain electromagnetic force. As shown in Fig. 7, radial component of electromagnetic force in the optimized shape by on/off method is greatly reduced. To confirm the effect of bubbles in the optimized model, comparison models as shown in Fig. 8 are prepared. Table 3 summarizes torque and displacement properties of them. We can see that the model (b) that have the bubbles in the circled area in Fig. 5(a) has the smallest displacement. Thus, this bubbles well work to reduce the displacement.

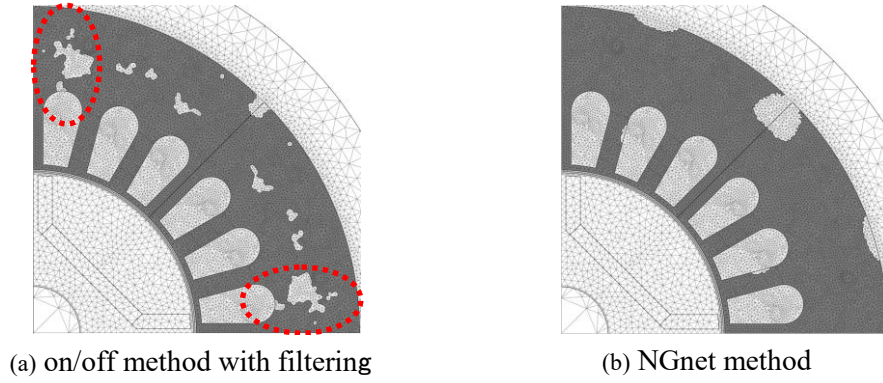


Figure 5: Optimal Shapes

Table 2: Comparison of Performance values.

Model	Average Torque [Nm]	Torque Ripple [Nm]	Average Displacement[nm]
Reference Model	2.315	1.162	53.67
Optimal Shape (with filtering)	2.270	0.9634	9.326
Optimal Shape (NGnet)	2.325	0.8161	41.88

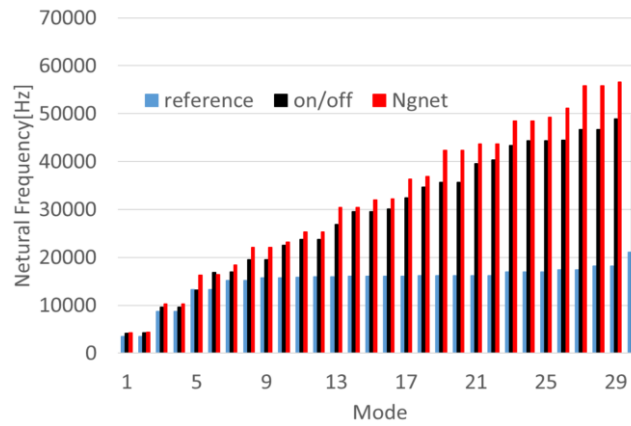


Figure 6: Comparison of natural frequencies.

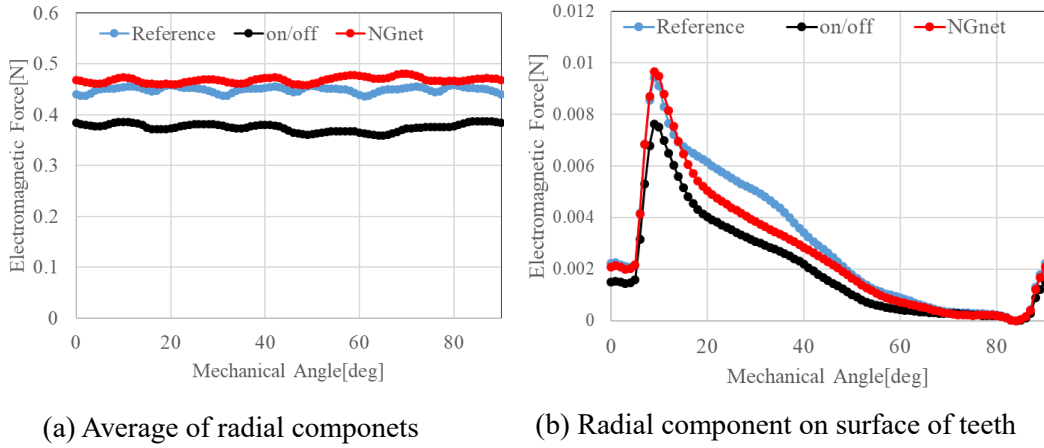


Figure 7: Comparison of time domain electromagnetic force.

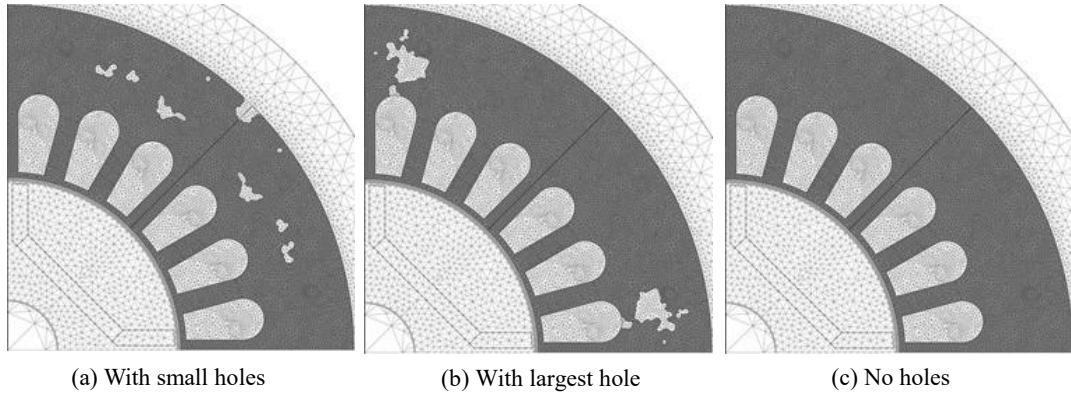


Figure 8: Comparison models for checking the effect of bubbles in shape optimized by on/off method

Table 3: Comparison of Performance values in comparison models in Fig .8

Model	Average Torque [Nm]	Torque Ripple [Nm]	Average Displacement [nm]
(a)	2.360	0.7086	41.89
(b)	2.271	0.9643	8.776
(c)	2.361	0.7089	39.38

4. Conclusion

In this study, topology optimizations to reduce the motor stator vibration are performed. To avoid the appearance of grayscale elements, the on/off method is employed. The conventional on/off method often yields complex shapes. This problem significantly degrades the performance of TO of motor. To solve this problem, we introduced two TO methods: the on/off method with filtering and NGnet method. The optimized geometry obtained by the proposed

method reduced the vibration while maintaining the torque performance. In the future, we will perform optimizations for different motor models to show the effectiveness of present methods.

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